

Audio Software Engineer & Researcher in the field of Computer Science, Human-Computer Interaction, Interactive System, Audio.

Author of 4 patents with publications & system work funded by *the U.S. National Cancer Institute, National Institutes of Health, Department of Defense*, and featured on medias (Forbes, NBCNews).

Interests & proven skills in *Programming, Audio Production, Sound Design, Game Engine, Middleware, Data Mining & Social Good*. An independent developer & a dedicated team player with interdisciplinary collaboration experiences.

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Google Scholar: <https://scholar.google.com/citations?user=Cj6CoIwAAAAJhl=en>

EDUCATION

University of Southern California, Los Angeles, CA

- MSc. in Computer Science. Topic: *Human—Computer Interaction, Data Mining, Signal Processing, Artificial Intelligence, Visualization, Bioinformatics*.

Ho Chi Minh City University of Technology, Vietnam National University, Ho Chi Minh City, Vietnam.

- Honor B.Eng. in Computer Science & Engineering. Topic: *Human—Computer Interaction, Human Biometrics, Data Privacy, Mobile Security*.

University of Washington, Seattle, WA

- Professional Cert. in Audio Engineering. Topic: *Audio Production in Music & Video Post-production*.

Other Certifications & Courses

- Virtual Acoustics in Immersive Audio, Stanford University CCRMA
- Deep Learning for Music Information Retrieval Workshop, Stanford University CCRMA
- Ableton Live Techniques: Synthesis and DSP, Berklee Online

SELECTED WORK EXPERIENCES

Lead Technical Sound Designer/Audio Programmer 2023–Present
Arrival (former as Formosa Interactive) Seattle, WA

- Designing, developing and optimizing audio systems & integration for game engines (Unreal Engine, Unity, and other proprietary game engines) and audio middleware (Wwise).
- Collaborating with 10+ cross-functional teams, including audio directors, audio leads, sound supervisors, sound designers, QA/Test, SWEs, and Production. Conducting bug fixes and code reviews for other engineers.
- Notable projects including :
 - Bungie's [Destiny 2](#) (released - AAA): Audio Integration/Implementation for The Final Shape [\[Trailer\]](#) & Season of the Wish [\[Trailer\]](#) (2023-2024).
 - Unannounced Project: Lead audio programming role (2024-2025).
 - Moon Studios' [No Rest for the Wicked](#) (released - [\[Trailer\]](#)): Audio programming & optimization (2025-Present).

Audio Programmer Volunteer / Independent Researcher 2020–Present

- Serving as a reviewer & program committee member for top-tier journal & conferences in the field of human-computer interaction, signal processing, audio, data mining & artificial intelligence: [CHI](#), [CSCW](#), [ICMI](#), [AM](#), etc.
- Serving as a volunteer audio programmer at [Games for Love](#) developing VR game acted as distraction therapy that has helped ease suffering for 500,000 children under treatment at hospitals & other child well-being facilities.
- Developing PC & mobile application to play spatial (ambisonics, binaural) audio using Google Resonance Audio, Oculus Audio Spatializer, Unity, C#.

Technical Audio Volunteer/Intern – San Francisco, CA Jan. 2021–May 2021
Women's Audio Mission (WAM)

- Teaching and technical supporting for 12 middle school students in 2 Music Coding classes using Python & EarSketch platform as part of [Girls on The Mic program](#).
- Video & audio production: dialogue, music, visuals for WAM's social content and events using Pro Tools, RX, After Effects.
- Technical supporting for WAM online streaming events (Girls on The Mic Showcase, WAMcon Virtual Nashville).

Research Fellow – Yorktown Heights, NY May 2017–Aug. 2017
IBM Research AI

- Publishing 3 patents & presenting at NeurIPS 2017, D4GX 2018 conferences. Media featuring on [Forbes](#), [NBCNews](#), [ACM](#), etc.
- Developing & evaluating the system of Simpler Voice: Overcoming Illiteracy - an AI natural language processing system for low-literate adults by decoding complex texts into simple messages, visual & audio elements.
- Working with 5 Research Scientists & Software Engineers at [IBM Research AI](#) and [NGO Literacy Coalition of Central Texas](#).

Research Assistant – Los Angeles, CA 2014–2018
USC Integrated Media Systems Center

- Publishing 4 papers. System demonstrated at the Former Vice President-President Joe Biden's [Cancer Moonshot](#) initiative: SXSL event at the U.S. White House: Press news [1] [2]. Presenting at ICDM 2015, AMIA 2016, ICDM 2017 conferences. Featuring on [Forbes](#), [NBC4News](#), [The Hill](#), etc.
- Developing, deploying, manual writing & technical supporting on-site/on-phone at 3 local LA hospitals with 55 patients for ATOM-HP: a system supporting physicians to quantify cancer patients' performance during chemotherapy by analyzing signals from wearables worn by patients & 3D camera sensors.
- Interdisciplinary research. The ATOM-HP program is a joint effort of the National Cancer Institute's Center for Strategic Initiatives and the Department of Defense's Rapid Response Technology Office. The project described was supported in part by [award number P30CA014089](#) from the National Cancer Institute.

Teaching Assistant & Guest Lecturer

2013, 2014–2018

University of Southern California (USC) – Los Angeles, CA*Ho Chi Minh City University of Technology (HCMUT)* – Vietnam

- Teaching assistant for 2 courses at USC: CSCI-585 Database Systems (2016, 2017) with 500+ undergrad/master students each class. Holding office hours for lessons review & tutorials, exam proctoring & grading.
- Guest lecturer for USC INF-552 Machine Learning for Data Informatics (2015).
- Teaching assistant for Artificial Intelligence (2013) at HCMUT. Teaching tutorial/lab sessions for 50+ students each session, preparing exams & grading.

Research Student – Ho Chi Minh City, Vietnam

2012–2014

Ho Chi Minh City University of Technology, Vietnam National University

- Publishing 1 paper at ICT-EurAsia 2014. Contributed to the Mobile Data Privacy project at D-STAR lab funded by Vietnam National University.
- Developing a mobile phone authentication framework (Android & iOS) using human biometrics: voice & facial recognition.
- Collaborating with 1 research scientist at D-STAR lab and 2 other research students.

Web Developer Intern – Ho Chi Minh City, Vietnam

May 2012–Sept. 2012

East Agile

- Participating in 2 projects: 1 internal and 1 client released project. Pair programming, Agile methodology.
- Developing, testing & deploying an internal Human Resources web application using Ruby, Ruby-on-Rails (RoR) framework.
- Project management & communicating with customer on the client project. Developing, testing, & deploying a video streaming web application.

PUBLICATIONS**Peer-reviewed Papers**

1. **Minh N.B. Nguyen***, Zaki Hasnain*, Ming Li, Tanya Dorff, David Quinn, Sanjay Purushotham, Luciano Nocera, Paul K. Newton, Peter Kuhn, Jorge Nieva, and Cyrus Shahabi. [Mining Human Mobility to Quantify Performance Status](#). *Demonstration Paper at the 2017 IEEE International Conference on Data Mining (ICDM)*, New Orleans, LA, USA, Nov. 18-21, 2017.
2. **Minh Nguyen**, Sanjay Purushotham, Hien To, and Cyrus Shahabi. [m-TSNE: A Framework for Visualizing High-Dimensional Multivariate Time Series](#). *The 2016 Workshop on Visual Analytics in Healthcare in conjunction with the American Medical Informatics Association (AMIA) Annual Symposium*. Chicago, IL, USA, Nov. 12-16, 2016.
3. Jiun-Yu Kao, **Minh Nguyen**, Luciano Nocera, Cyrus Shahabi, Antonio Ortega, Carolee Winstein, Ibrahim Sorkhoh, Yu-chen Chung, Yi-an Chen, and Helen Bacon. [Validation of Automated Mobility Assessment Using a Single 3D Sensor](#). *The 4th International Workshop on Assistive Computer Vision and Robotics in conjunction with the 14th European Conference on Computer Vision (ECCV)*. Amsterdam, The Netherlands, Oct. 9th, 2016.

4. **Minh Nguyen**, Liyue Fan, and Cyrus Shahabi. [Activity Recognition Using Wrist-Worn Sensors for Human Performance Evaluation](#). *The 6th Workshop on Biological Data Mining and its Applications in Healthcare in conjunction with the 14th IEEE International Conference on Data Mining (ICDM)*. Atlantic City, New Jersey, USA, Nov. 14-17, 2015.
5. Van N. Nguyen, Vuong Q. Nguyen, **Minh N.B. Nguyen**, and Tran K. Dang. [Fuzzy Logic Weight Estimation in Biometric-Enabled Co-authentication Systems](#). *The 2014 Information and Communication Technology - EurAsia Conference (ICT-EurAsia)*. Bali, Indonesia, Apr. 14-17, 2014.

Posters

1. **Minh N.B. Nguyen**, Samuel Thomas, Anne E. Gattiker, Sujatha Kashyap, and Kush R. Varshney. [SimplerVoice: A Key Message & Visual Description Generator System for Illiteracy](#). *The 2018 Data for Good Exchange Conference (D4GX)*. New York, NY, USA, Sept. 16, 2018.
2. **Minh N.B. Nguyen**, Samuel Thomas, Anne E. Gattiker, Sujatha Kashyap, and Kush R. Varshney. [SimplerVoice: Overcoming Illiteracy](#). *The 2017 Neural Information Processing Systems - Women in Machine Learning Workshop*. Long Beach, CA, USA, Dec. 7, 2017.
3. **Minh N.B. Nguyen**, Mehrnoosh Mirtaheri, Joanne Kao, Aria Azadegan, Luciano Nocera, Sanjay Purushotham, Peter Kuhn, Jorge Nieva, and Cyrus Shahabi. [ATOM-HP: Analytical Technologies to Objectively Measure Human Performance - Data Analysis](#). *The 2017 USC Integrated Media Systems Center retreat*. Los Angeles, CA, USA, Mar. 23, 2017.
4. **Minh N.B. Nguyen**, Luan Tran, Kien Nguyen, Aaron Mejia, Luciano Nocera, Liyue Fan, Cyrus Shahabi, Peter Kuhn, and Jorge Nieva. [ATOM-HP: Analytical Technologies to Objectively Measure Human Performance - System & Data Preprocessing](#). *The 2016 USC Integrated Media Systems Center retreat*. Los Angeles, CA, USA, June 1, 2016.
5. **Minh N.B. Nguyen**, Liyue Fan, and Cyrus Shahabi. [Human Activity Recognition with Wearable Devices](#). *The 2015 USC Integrated Media Systems Center retreat*. Los Angeles, CA, Apr. 2, 2015.

Technical Reports

1. **Minh N.B. Nguyen**, Samuel Thomas, Anne E. Gattiker, Sujatha Kashyap, and Kush R. Varshney. [SimplerVoice: A Key Message & Visual Description Generator System for Illiteracy](#). *The 2018 Data for Good Exchange Conference (D4GX)*. New York, NY, USA, Sept. 16, 2018.
2. Hien To, Sasan Tavakkol, Seon H. Kim, **Minh Nguyen**, and Cyrus Shahabi. [On Acquisition and Analysis of Visual Data for Crowdsourcing Disaster Response](#). *Technical Report, University of Southern California*, 2016.

PATENTS

1. Anne E. Gattiker, Sujatha Kashyap, **Minh N. B. Nguyen**, Samuel Thomas, Kaipeng Li, Thomas Hubregtsen. [Constructing, evaluating, and improving a search string for retrieving images indicating item use](#). *Patent No. 11645329*. United States, May 09, 2023.

2. Sujatha Kashyap, Anne E. Gattiker, Kaipeng Li, Samuel Thomas, **Minh N. B. Nguyen**, and Thomas Hubregtsen. [Constructing, evaluating, and improving a search string for retrieving images indicating item use](#). *Patent No. 11061943*. United States, July 13, 2021.
3. Anne E. Gattiker, Samuel Thomas, **Minh N. B. Nguyen**, Sujatha Kashyap, and Thomas Hubregtsen. [Constructing, evaluating, and improving a search string for retrieving images indicating item use](#). *Patent No. 11055345*. United States, July 6, 2021.
4. Sujatha Kashyap, Anne E. Gattiker, Kaipeng Li, Samuel Thomas, **Minh N. B. Nguyen**, and Thomas Hubregtsen. [Constructing, evaluating, and improving a search string for retrieving images indicating item use](#). *Patent No. 10664517*. United States, May 26, 2020.

PROFESSIONAL & COMMUNITY ACTIVITIES

- **Program Committee Member** of [The 7th Workshop on Mining and Learning from Time Series \(MiLeTS\)](#) in conjunction with the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD 2021).
- **Scientific Committee Member** of [The 2021 International Audio Mostly Conference: A Journey in Sound \(AM'21\)](#).
- **Reviewer** of [The 2024 ACM CHI Conference on Human Factors in Computing Systems \(CHI 2024\)](#).
- **Reviewer** of [The 2023 International Symposium on Mixed and Augmented Reality \(ISMAR 2023\)](#).
- **Reviewer** of [The 2022 ACM CHI Conference on Human Factors in Computing Systems \(CHI 2022\)](#).
- **Reviewer** of [2022 ACM International Joint Conference on Pervasive and Ubiquitous Computing \(UbiComp / ISWC\)](#).
- **Reviewer** of [The 14th ACM SIGCHI Symposium on Engineering Interactive Computing Systems \(EICS 2022\)](#).
- **Reviewer** of [The 23rd ACM International Conference on Multimodal Interaction \(ICMI 2021\)](#).
- **Reviewer** of [The 24th ACM Conference on Computer-Supported Cooperative Work and Social Computing \(CSCW 2021\)](#).
- **Reviewer** of [ACM Transactions on Spatial Algorithms and Systems \(TSAS\)](#).
- **Member/Student Member** of The Audio Engineering Society [AES](#), [IEEE](#), [Systers](#), [SoundGirls.org](#).
- **Teaching/Technical Supporting** 12 [Aim High](#) middle school students in 2 classes of Audio Coding in the Girls on the Mic program of [Women's Audio Mission](#) 2021.
- **Live Audio Volunteer** at [The Vera Project](#) 2020.
- **Mentoring** 3 high school students and 10+ undergrad/master students at USC in ATOM-HP project 2014-2017.
- **Student Volunteer** at [Grace Hopper](#) 2016, [ICDM](#) 2015, [SSEAYP](#) 2013.

HONORS & AWARDS

- Stanford's CCRMA Summer Workshops Scholarship 2024, 2025.
- Game Audio Diversity Alliance Scholarship 2021.
- SoundGirls' Leslie Ann Jones Scholarship in Honor of Ethel Gabriel 2020.
- IBM Social Good Fellowship 2017.
- Computer Research Association - Women Travel Award Grad Cohort 2017.
- USC Women in Science and Engineering Travel Grant for IEEE ICDM 2017.
- GSG Travel Grant for AMIA VAHC 2016.
- USC Grace Hopper Scholarship 2016.
- Student Travel Award for IEEE ICDM 2015.
- Vietnam Education Foundation Fellowship 2014-2017.
- HCMUT Dean's Excellent Student (Top 1%) Award B.Eng. 2014.
- Kanden-SS Scholarship 2013.
- Odon Vallet Scholarship 2013.
- HCMUT Dean's Honor Scholarships covering B.Eng. full tuition 2009 - 2014.

PODIUM PRESENTATION & TALKS

- System demonstration at [ICDM 2017 - the IEEE International Conference on Data Mining](#). "*Mining Human Mobility to Quantify Performance Status*", New Orleans, LA, USA, Nov. 19, 2017.
- Podium presentation at [AMIA 2016 - American Medical Informatics Association Annual Symposium](#). "*m-TSNE: A Framework for Visualizing High-Dimensional Multivariate Time Series*", Chicago, IL, USA, Nov. 12, 2016.
- Podium presentation at [ICDM 2015 - the IEEE International Conference on Data Mining](#). "*Activity Recognition Using Wrist-Worn Sensors for Human Performance Evaluation*", Atlantic City, NJ, USA, Nov. 14, 2015.
- Guest lecturer at [Informatics 552 - Machine Learning for Data Informatics](#). "*Activity Recognition with Wearables*", University of Southern California, Fall 2015.

MEDIA

- Project **ATOM-HP** was featured on:
 - ScienceBlog. "[Wearable Fitness Trackers Can Help Doctors Treat Cancer Patients](#)" (Feb. 15, 2021).
 - DailyTrojan. "[USC researchers tackle medical issues at the Michelson Center](#)" (May 29, 2018).
 - HuffPost. "[How Can Wearable Technology Improve Cancer Treatment?](#)" (Dec. 6, 2016).

- Forbes. *"Wearable Technology Is Improving Cancer Treatment One Day At A Time"* (Oct. 19, 2016).
- The Hill. *"Technology can be used to bridge the gap between doctor's visits"* (Sept. 30, 2016).
- NBC4News. *"Wearable Tech Improves Cancer Treatment"* (Sept. 29, 2016).
- USC News. *"USC to show how wearable technology can improve cancer treatment as part of White House event"* (Sept. 28, 2016).
- ScienceDaily. *"How wearable technology can improve cancer treatment"* (Sept. 28, 2016).
- National Cancer Institute, Center for Strategic Scientific Initiatives. *"Analytical Technologies to objectively measure human performance (ATOM-HP)"*.
- Project **SimplerVoice: Overcoming Illiteracy** was featured on:
 - Forbes. *"10 Wonderful Examples Of Using Artificial Intelligence (AI) For Good"* (Jun. 22, 2020).
 - CIO. *"How AI/ML is helping to eradicate poverty"* (Jun. 18, 2019).
 - ACM News. *"Can AI Solve Poverty?"* (Sept. 5, 2017).
 - THINK Blog. *"Clarifying the Complex with a 'Simpler Voice'"* (Jul. 31, 2017).
 - NBCNews. *"AI Is a Game-Changer in the Fight Against Hunger and Poverty. Here's Why"* (Jun. 21, 2017).
 - Futurism. *"IBM's Newest Program Uses AI to Solve the Biggest Problems Facing Humanity Today"* (Jun. 9, 17).
 - Mashable. *"IBM turns to artificial intelligence to solve poverty, hunger, and illiteracy"* (Jun. 7, 2017).

TECHNICAL SKILLS

- *Programming:* C/C++, C#, Python, Java, MATLAB, Max/MSP, JavaScript, Ruby, Scala, HTML, CSS.
- *Digital Signal Processing/AI tool:* NumPy, Matplotlib, NLTK, Weka, TensorFlow, PyTorch.
- *Digital Audio Workstation:* Pro Tools, Ableton Live.
- *SDK:* JUCE, Google Resonance Audio, Kinect.
- *Game Engine & Middleware:* Unreal Engine, Unity, Wwise (certified), FMOD.
- *Database:* MySQL, Postgres, NoSQL, Spatial-DBs, MapReduce.
- *Platform:* PC, Console (PS5, Xbox, Switch2), Mobile (Android, iOS, Web).
- *Version Control:* Perforce, SVN, Git
- *Others:* VR / AR, Agile methodology, Technical Sound Design, Recording Mixing, Interdisciplinary & Collaborative research.

REFERENCES Available upon request.